

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Artificial Intelligence

2. Subject Name: Statistical Foundations of Artificial Intelligence and Machine Learning L-T-P: 3-1-0 Credits: 4

3. Pre-requisites:

4. Syllabus and reference books:

Syllabus:

Introduction: Statistics in AI -ML Problems, Univariate, multivariate, sampling distributions, Bayesian set-up of distributions, sampling methods including MCMC. Parametric Inference: Estimation and Testing of hypothesis with LLN, Mixture model with EM. Linear model with regularization, Generalized linear model with the Bayesian concept. Resampling techniques and model validation. Nonparametric Inference: Kernel density estimation, the goodness of fit tests, Gaussian process, functional CLT and function estimation, Variational inference

Reference Books:

1. Kevin Murphy (2012). Machine Learning: A Probabilistic Perspective. MIT Press
2. Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.

Others:

1. Hastie, T., Tibshirani, R., Friedman, J., & Franklin, J. (2005). The elements of statistical learning: data mining, inference and prediction. The Mathematical Intelligencer, 27(2), 83-85.
2. Kulkarni, S., & Harman, G. (2011). An elementary introduction to statistical learning theory (Vol. 853). John Wiley & Sons.
3. Herrera, C. N. B. (2013). Modern Multivariate Statistical Techniques: Regression, Classification, and Manifold Learning. Investigación Operacional, 30(2), 190-192.
4. Vapnik, V. (2013). The nature of statistical learning theory. Springer science & business media.
5. Duda, R. O., Hart, P. E., & Stork, D. G. (2012). Pattern classification. John Wiley & Sons.

5. Lecture-wise break-up:

SI No.	Module	Unit	Module Hr
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			(48)
1	Introduction and Motivation	Statistics in AI -ML Problems: Topic Modeling, Image/video analysis	2
2	Probability Distributions	(a) Axiomatic definition of probability (b) Binomial, Poisson, Gamma, Beta, Dirichlet, Gaussian, Mixture Gaussian (c) Multivariate Gaussian, Distribution of quadratic forms, Chi-squared, T, F (d) Moment generating functions	6
3	Parameter Estimation	(a) Parametric estimation :Method of Moment estimator, Maximum likelihood estimator, Bayesian Estimation and conjugate priors (b) Unbiasedness, Law of large numbers: Consistency	4
4	Sampling Methods	(a) Rejection sampling , importance sampling (b) Markov Chain and its convergence/central limit theorem (c) Metropolis-Hastings algorithm	4
5	Mixture Models and EM	(a) EM Algorithm with convergence (b) Mixture Gaussian	4
6	Testing of Hypothesis	(a) Hypothesis, Errors, power (b) One and two sample normal test (c) Permutation Test, A/B testing	4
7	Probabilistic Modeling of Linear Regression	(a) Multiple linear regression, Polynomial regression	4

		with the least square condition, autoregression. (b) Regularisation : Lasso, Ridge, principle component regression (c) Estimation and testing of regression coefficients.	
8	Generalized Linear Models for Classification	(a) Logistic regression (b) Linear & quadratic discriminant analysis (c) Bayesian Logistic regression	4
9	Model Selection	(a) Cross-validation, validation set (b) Jack-knife/ Bootstrap (c) AIC / BIC for model selection	4
10	Non-parametric Inference	(1) Kernel density estimation (2) Empirical CDF & Kolmogorov-Smirnov test (non-parametric) (3) Chi-squared test for goodness of fit (categorical)	4
11	Topic Model Analysis	(a) Using Gibbs sampling for approximate inference (b) Using Variational Inference	4
12	Extreme Value Theory	a. Peak-over-Threshold and Block-maxima b. GEV, Generalized Pareto distributions, Q-Q plots c. Applications in Geosciences	